LESSON 6.6

# Identify Equivalent Algebraic Expressions

**Start here** · Words, worked example, and sentence starters

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can show that two expressions are equivalent by simplifying and combining like terms.

**WHY THIS MATTERS** At the music studio, two producers wrote formulas for the cost of renting equipment. Producer A wrote 3x + 2x + 10, and Producer B wrote 5x + 10. They look different, but are they really? Let's check: if x = 4, Producer A gets 3(4) + 2(4) + 10 = 30, and Producer B gets 5(4) + 10 = 30. Same result — they're equivalent expressions!

START HERE

## Words for this lesson

Read each word before you begin. Say it out loud.

| WORD | WHAT IT MEANS | EXAMPLE |
|---|---|---|
| **Equivalent Expressions** *Spanish: Expresiones equivalentes* | Expressions that look different but have the same value for every allowed variable value. | 7c + 9 + 6c = 13c + 9 |
| **Equivalent** *Spanish: Equivalente* | Expressions that always have the same value. | 2x + 4 and 2(x + 2): when x = 3, both = 10; when x = 7, both = 18 — always the same |
| **Simplify** *Spanish: Simplificar* | To rewrite an expression in a shorter form that is still equal to the original. | 7c + 9 + 6c → 13c + 9 |
| **Like Terms** *Spanish: Términos semejantes* | Terms with the same letter raised to the same power, like 2x and 5x. | 7c and 6c |
| **Combine** *Spanish: Combinar* | To add or subtract terms with the same letter into one. | 7c + 6c = 13c |
| **Coefficient** *Spanish: Coeficiente* | The number in front of a letter, like the 3 in 3x. | the 13 in 13c |

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**Simplify 7c + 9 + 6c. Is it equivalent to 13c + 9?**

1. Like terms: 7c and 6c.
2. Combine: 7c + 6c = 13c. The 9 stays alone.
3. 7c + 9 + 6c = 13c + 9.

**Answer:** 13c + 9; yes, equivalent

NOW YOU TRY

## Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

**Let's check 2x + 3x and 5x. Try x = 2.**

1. _____
2. _____
3. _____

**Answer:** _____

SAY IT AND WRITE IT

## Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The like terms are _____ and _____ because _____ .
- I combined them to get _____ .
- The constant _____ stays by itself because _____ .
- The two expressions _____ equivalent because _____ .

**Word bank** equivalent | like terms | combine | simplify | coefficient | constant | term | variable | same value | group | add | compare

WATCH OUT!

## A common mistake

A common mistake in Equivalent Expressions is adding the exponents instead of the coefficients when combining like terms

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# Identify Equivalent Algebraic Expressions

**Version A** · Supported practice

**MASTERY CHECK**
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

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**WHY THIS MATTERS** At the music studio, two producers wrote formulas for the cost of renting equipment. Producer A wrote 3x + 2x + 10, and Producer B wrote 5x + 10. They look different, but are they really? Let's check: if x = 4, Producer A gets 3(4) + 2(4) + 10 = 30, and Producer B gets 5(4) + 10 = 30. Same result — they're equivalent expressions!

## PART 1 Understand the idea

models and meaning

**1** **Circle the letter of the best answer. Show how you know.**

Which expression is equivalent to 4x + 3x?

(A) 7x
(B) $7x^2$
(C) 12x
(D) 43x

**HINT** 4x and 3x share the exact same variable, x — that makes them like terms.

SHOW YOUR WORK

**2** **Circle the letter of the best answer. Show how you know.**

Which expression is equivalent to 2(m + 5)?

(A) 2m + 10
(B) 2m + 5
(C) 7m
(D) m + 10

**HINT** The 2 outside the parentheses must multiply both the m and the 5.

SHOW YOUR WORK

**3** **Write the name of the correct group on each line.**

Sort each pair of expressions by whether they are equivalent or not.

**Groups:** **Equivalent** **Not Equivalent**

_____ 3x + 2x and 5x

_____ 2(n + 4) and 2n + 8

_____ 4a + 3 and 7a

_____ x + x and 2x

_____ 3y + 2 and 5y

**HINT** Equivalent means always equal. Simplify the first expression in each pair and compare with the second.

## PART 2 Apply it

real-world problems

**4** **Circle the letter of the best answer. Show how you know.**

Which property is shown: 3(x + 7) = 3x + 21?

(A) Commutative

(B) Distributive

(C) Associative

(D) Identity

**HINT** Watch what happened to the 3: it multiplied the x AND the 7 across the parentheses.

**SHOW YOUR WORK**

**5** **Circle the letter of the best answer. Show how you know.**

Simplify: 5n + 2n

(A) 7n

(B) $7n^2$

(C) 10n

(D) 52n

**HINT** 5n and 2n both have the variable n — they're like terms, so they can combine.

**SHOW YOUR WORK**

## PART 3 Explain and correct

reasoning

**6** **Find the mistake. Explain it, then show the correct work.**

### Spot the Distributive Mistake

1. **Expression** 3(x + 4)
2. **Distribute** Multiply the 3 by the first term only: 3 · x + 4
3. **Result** 3(x + 4) = 3x + 4

**Which step has the mistake? Explain what went wrong, then show the correct work.**

**HINT** How many terms are inside the parentheses? The 3 multiplies each one.

CORRECT WORK

**Correct answer:** _____

EXPLAIN YOUR THINKING

**How did you decide which expressions belong in the same group? What strategies did you use?**

**SENTENCE STARTER** I knew ___ and ___ are equivalent because when I ___ the terms, I got the same expression.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.6

# Identify Equivalent Algebraic Expressions

**Version B** · Core practice

**MASTERY CHECK**
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can show that two expressions are equivalent by simplifying and combining like terms.

**WHY THIS MATTERS** At the music studio, two producers wrote formulas for the cost of renting equipment. Producer A wrote 3x + 2x + 10, and Producer B wrote 5x + 10. They look different, but are they really? Let's check: if x = 4, Producer A gets 3(4) + 2(4) + 10 = 30, and Producer B gets 5(4) + 10 = 30. Same result — they're equivalent expressions!

## PART 1 Understand the idea

models and meaning

**1** **Write the name of the correct group on each line.**

Match each expression with its simplified form.

**Groups:** Correctly Matched

_____ 8y + 2y → 10y

_____ 3(a + 6) → 3a + 18

_____ 5n + 4 + 2n → 7n + 4

_____ 9p − 3p → 6p

_____ 4(2x − 1) → 8x − 4

_____ 6t + t → 7t

**2** **Write the letter of the matching item on each line.**

Match each expression to its simplified equivalent.

_____ 6x + 2x

_____ 3(n + 5)

_____ 4a + 7 + a

_____ 2(3y − 4)

_____ 10m − 4m + 1

_____ 5(x + 2)

(A) 8x

(B) 3n + 15

(C) 5a + 7

(D) 6y − 8

(E) 6m + 1

(F) 5x + 10

**3** **Complete the table. Show how you found each value.**

Simplify each expression. Then evaluate both the original and simplified form when x = 3 to verify they are equivalent.

| Original | Simplified | Value when x = 3 |
|---|---|---|
| 2x + 5x + 1 | | |
| 4(x + 2) | | |
| 3x + 4 + x − 1 | | |

SHOW YOUR WORK

**4** **Decide whether each pair balances. Check Yes or No, then explain.**

Both equal 14 — they are equivalent because 3x + 2x = 5x.

| Left side | Right side | Balanced? |
|---|---|---|

EXPLAIN YOUR CHOICE

## PART 2 Explain and correct

reasoning

**5** **Find the mistake. Explain it, then show the correct work.**

### Spot the Combining Mistake

1. **Expression** 7x + 3 + 2x
2. **Combine** Combine every number you see: 7x + 2x + 3 = 9x + 3, then add the 9 and 3 too
3. **Result** 7x + 3 + 2x = 12x

**Which step has the mistake? Explain what went wrong, then show the correct work.**

CORRECT WORK

**Correct answer:** _____

EXPLAIN YOUR THINKING

**How did you decide which expressions belong in the same group? What strategies did you use?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.6

# Identify Equivalent Algebraic Expressions

**Challenge** · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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## PART 1 Understand the idea

models and meaning

**1** **Complete the table. Show how you found each value.**

For each expression, find an equivalent expression using a DIFFERENT method (distributive property, combining like terms, or factoring).

| Expression | Equivalent Form | Method Used |
|---|---|---|
| 4x + 8 | | |
| 3a + 2a + 5 | | |
| 6(n − 2) | | |

SHOW YOUR WORK

## PART 2 Explain and correct

reasoning

**2** Find the mistake. Explain it, then show the correct work.

### Find the Simplification Error

1. **Expression** $3x + 5 + 2x + 1$
2. **Combine** $3x + 2x + 5 + 1$
3. **Simplify** $5x^2 + 6$

**Which step has the mistake? Explain what went wrong, then show the correct work.**

CORRECT WORK

**Correct answer:** _____

**3** Answer in complete sentences.

A student says $2x + 3$ and $5x$ are equivalent because both equal 5 when $x = 1$. Is the student correct? Explain why testing one value is not enough to prove equivalence, and find a value of x where they give different results.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

**How did you decide which expressions belong in the same group? What strategies did you use?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help